

PAPER CODE NO.
COMP232

EXAMINER : Alexei Lisitsa
DEPARTMENT : Computer Science Tel. No. 0151 795 4250



U N I V E R S I T Y O F
L I V E R P O O L

Mock Exam Paper

CYBERSECURITY

TIME ALLOWED : Two Hours

INSTRUCTIONS TO CANDIDATES

Answer FOUR out of FIVE questions.

If you attempt to answer more questions than the required number of questions (in any section), the marks awarded for the excess questions answered will be discarded (starting with your lowest mark).

Answer four out of five questions listed.

1. (a) What are potential problems with the security protection based on the secrecy of an encryption algorithm?

(5 marks)

(b) A simple way of finding a collision in a hash function H is as follows

1. Pick initial value I and calculate $H(I)$;
2. Pick randomly further value J and compare $H(J)$ to $H(I)$;
3. Repeat step 2 until $H(J) = H(I)$.

Suggest an improvement of this attack, which would likely lead to faster discovery of the collision.

(10 marks)

- (c) Describe the Diffie-Hellman (DH) Key Exchange algorithm and the attack on this algorithm that uses the computation of discrete logarithms. Why is DH considered secure?

(10 marks)

2. (a) Define passive security attacks. Explain how the development of quantum cryptography may change the way of dealing with passive attacks.

(8 marks)

(b) Explain why RSA can be considered as a partially-homomorphic encryption algorithm.

(7 marks)

- (c) What is wrong in the following algorithm for computing a hash function? Take a message M , generate a random private RSA key K . Encrypt M with K and take the first 240 bits of the result as a hash of M .

(10 marks)

3. (a) Explain what are the advantages and disadvantages of using quantum generator of random numbers in cryptography

(5 marks)

(b) Describe the RSA-based blind signature technique. Discuss its role in the implementation of electronic payment systems and electronic voting systems.

(10 marks)

(c) Explain what is a zero-knowledge proof (ZKP). What are the possible applications of ZKP?

(10 marks)



4. (a) What is the common factor attack on RSA and why is it practically possible?

(8 marks)

(b) What are the two main categories of intrusion detection methods? Describe advantages and disadvantages of methods from each category.

(7 marks)

(c) A simple key exchange protocol shown below can be used to establish a key over an open channel. Here K_b is a public key of B, N_a and N_b are random numbers (nonces) chosen by A and B, respectively, and k_{ab} is the key for symmetric encryption to be exchanged.

- Message 1. $A \rightarrow B : N_a$
- Message 2. $B \rightarrow A : N_b$
- Message 3. $A \rightarrow B : \{k_{ab}, A, N_b\}_{K_b}$
- Message 4. $B \rightarrow A : \{N_a\}_{k_{ab}}$

What is the possible issue with this protocol? How can it be fixed?

(10 marks)

5. (a) For a block cipher, what is the Cipher Block Chaining mode (CBC)? What is the main purpose of using CBC mode? What is the main disadvantage of this mode?

(7 marks)

(b) What is a multifactor authentication technique? Give an example and explain the rationale behind it.

(8 marks)

(c) What is Fully Homomorphic Encryption (FHE)? What is the main issue with applications of FHE? Give an example of an application which would benefit from using FHE.

(10 marks)